

Step Smart

Injury Preventive Insoles



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PROBLEM AND SOLUTION

Many athletes suffer injuries due to improper foot posture, leading to inflammation, pain, and setbacks. Some injuries may force athletes to pause or even end their training or careers. According to the American Academy of Orthopedic Surgeons, 25% of sports injuries are foot and ankle related. We aim to design a solution that corrects poor placement and prevents injuries before they happen. By tracking foot placement and pressure in real time through sensors in insole inserts and connecting that data to a phone app, athletes could correct their movement before injury happens. Thin inserts with sensors on main pressure points allow for comfort while keeping typical movement undisturbed like chunky inserts might do. The data is displayed on the app in readable and understandable graphs and tables for individual users as well as coaches.

INITIAL PAPER PROTOTYPE

Insole Insert



Figure 1. Initial Insert Paper Prototype

The insert comes in a variety of sizes to account for different shoe types such as ballet slippers or soccer cleats. Sensors are on the main parts of the foot to read pressure placement to calculate over pronation vs supination. They are made to be thin and flexible so they are

inconspicuous once in the shoe.

App

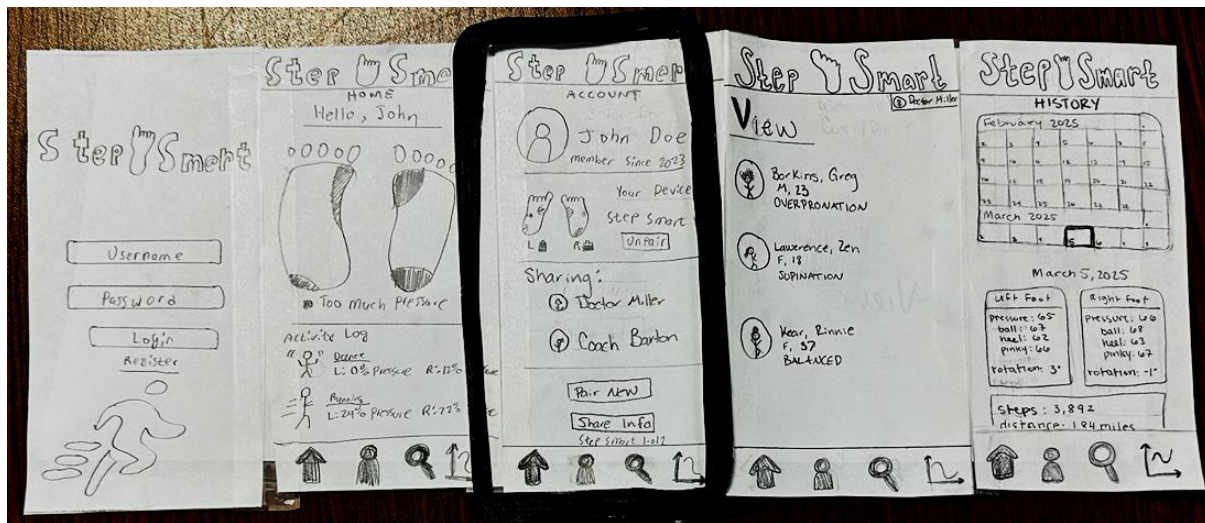


Figure 2. Initial App Paper Prototype

The app includes multiple screens for a variety of functionalities. The login screen allows you to register and log in to the app. Once logged in, the home screen offers a real time foot positioning display as well as an activity log. The account screen is there to view your device list and sharing list as well as your typical account info. The view page is for coaches, instructors, trainers, etc. to view the team members of those you are in charge of so you can see how their feet are positioned while wearing the device. Finally, the history page allows the user to track foot positioning over time in a calendar format.

Tasks

Know How One's Feet Are Positioned During Activities

- When the user puts the insoles in their shoe during an activity, they or their coach/instructor can open the app either during or after the activity to view their data. The foot positioning data is synced to the app in real time and offers easy to understand maps and charts of pressure to view in order to reposition their foot accordingly.

Use an App to See Health and Fitness Data Visuals

- Once the user is logged into our intuitive app, they can go to the home page, history page, or view page (if a coach) to view their data in different forms. The home tab offers a clear visualization map of the foot to see where pressure is being placed on the foot, so the user can reposition if needed. An activity log is also kept on the home page as a clear and readable chart. The history tab allows the user to see detailed information on

their posture trends as well as view the specific data on a chosen day in the calendar. The view tab is for coaches and instructors to view the data of their students and team members, so they can correct the student's feet while wearing the device.

TESTING PROCESS

Methods

The test was divided into two main parts. First, the prototype insoles were tested in the participants shoes to assess comfort, fit, and suitability for different types of footwear. The participant moved around in the prototype and provided initial impressions and feedback. The second portion focused on the app. The participant was initially given free rein to explore the interface, sharing their thought processes as they looked through the screens highlighting which features they would use most frequently. This was followed by more structured tasks, where we prompted the participants to locate specific features and describe what kind of visualizations or metrics they would expect to see.

Participants

Participant 1: Our first usability test was conducted with Jayme's sister, a high school senior with 15 years of dance experience and 5 years of volleyball experience. Her background in athletics and her strong interest in health and fitness made her an ideal candidate for testing both the device and app. The test took place in a spacious room at the participant's home, allowing for unrestricted movement. This familiar and comfortable environment encouraged open feedback and allowed for clarifying questions throughout the session, creating a more authentic and informative user experience.

Participant 2: Our second usability test was conducted by a computer science classmate. He is in his 20s and works out regularly. He also has a history of different sports before college such as track, swimming, etc. His background in activity and technology development makes him an ideal candidate for testing the prototype. The test took place after a class he, Jayme, Evan, and Sophie had together for convenience for the participant and the team members. It was done in the Fowler Student Design Center's conference room in the Engineering Building, which was a secluded and comfortable environment to allow for complete assessment without distractions or unease.

Participant 3: Our third usability test was conducted by a computer science classmate. He is in his 20s and has a history of knee issues, which could stem from improper foot placement. His background in technology development makes him a great candidate for testing the prototype,

especially the app for more input in his ideas for the user interface. The test took place after a class that he, Jayme, Evan, and Sophie had together for convenience for the participant and the team members. It was done in the conference room in the Engineering Building, which was a secluded and comfortable environment to allow for complete assessment without distractions or unease.

Refining Process

Since our first test was done with one group member, we immediately recognized the importance of having more team members present during testing to improve the flow, capture more detailed observations, and ensure a more structured approach to questioning, which is what we did for the next two tests. Also, for the second two tests, we heavily modified our prototype app to add in the features that were mentioned from the first test, and the insert was modified to add grip. Based on these changes, we had to update some of our questions. We also knew to base some of the questions on the issues from the first test by adding more emphasis on certain utilities to see if we fixed the issues.

TEST RESULTS

Our paper prototype was subject to three different types of evaluations: an initial design critique and a heuristic evaluation conducted among other groups, and three usability tests conducted with individual participants.

The initial design critique proved to be tremendously useful for focusing our design's purpose and audience. After pitching our idea, we decided to focus on accessibility, prevention, and continual improvement rather than treating severe cases. From this critique we also expanded the design to be usable by medical professionals, not just coaches.



Figure 3. Paper Prototype Before Heuristic Evaluation

The majority of the feedback we received on the heuristic evaluation was related to user experience. Group 5 suggested we add a help page, a settings page, and additional buttons for data sharing, adding users on the View page, a button to sign out, and back buttons to return to the previous page. They also suggested we add an indicator to the tab bar to show the currently active page. Group 6 suggested that the calendar on the History page show daily posture improvement/deterioration at a glance. Their other app suggestions included urgent posture notifications, scrollability indication, and a reward/achievement system. For the insert, they suggested adding an extra sensor for more precise tracking and protection above the sensors themselves.

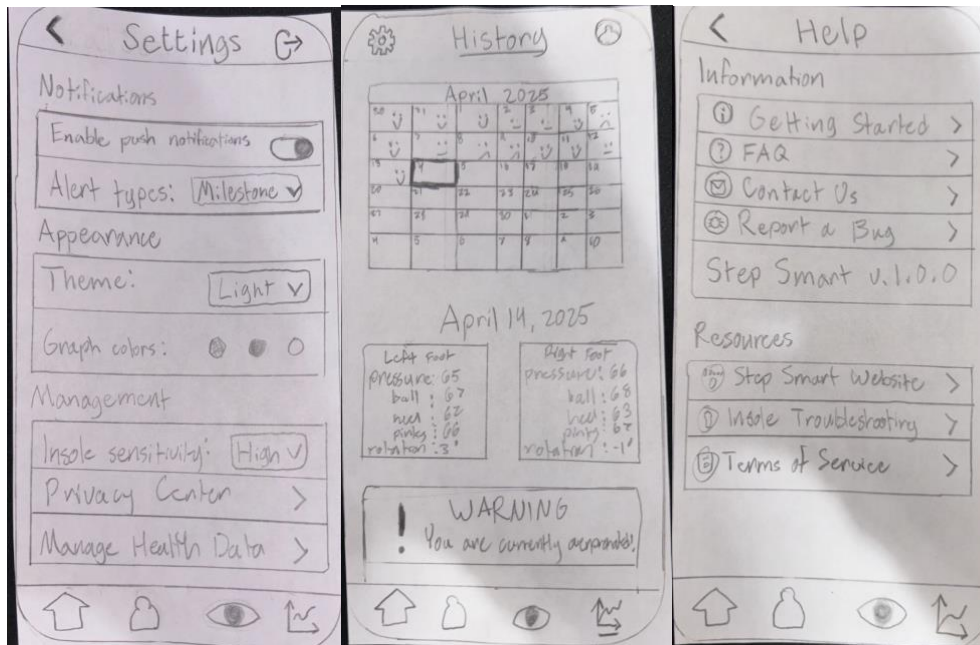


Figure 4. Paper Prototype After Heuristic Evaluation

In response to our heuristic evaluation, we made numerous revisions to improve our app's usability. We added a settings page with options for notifications, alert types, app themes, insole sensitivity, and data management. The settings page also contains a button to sign out. We added a help page with a getting started guide, FAQ, and links to troubleshooting and other resources. We added feedback to the History page's calendar as well as notifications for dangerous foot posture. Across the whole app we changed the top bar to show the name of the current page and added an underline to the tab bar to indicate the current tab.

Usability testing was the second method of evaluation that our paper prototype faced. We conducted a single test with one participant, making changes in accordance with their feedback before we conducted our final two tests. The participant said the insert was slippery, so rubber grips were added to the underside to hold it in place within the shoe. The purpose of the View page was unclear, so an option to toggle "coach mode" was added to separate the two different experiences. The icon for the page was also changed from a magnifying glass to an eye.

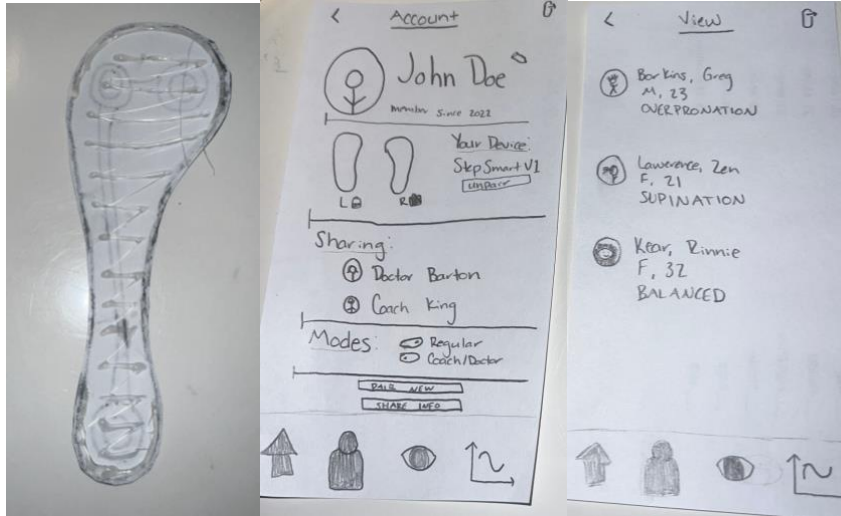


Figure 5. Paper Prototype After Usability Test 1

Finally, we conducted our second round of usability testing with the modified prototype made after round one. Participants two and three found that the added rubber grips secured the insert nicely. The purpose of the View page was still unclear, so it was overhauled into the Teams page, featuring a new list of existing teams so coaches and doctors can manage multiple groups at a time. The icon was also changed again from an eye to a list. The settings button was moved from the History page to the Account page, where a button to access the Help page was also added.

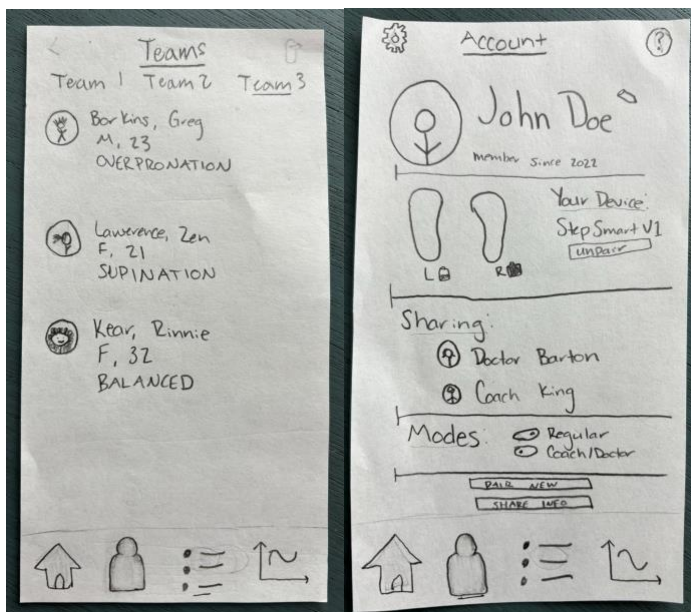


Figure 6. Paper Prototype After Usability Test 2

FINAL PAPER PROTOTYPE

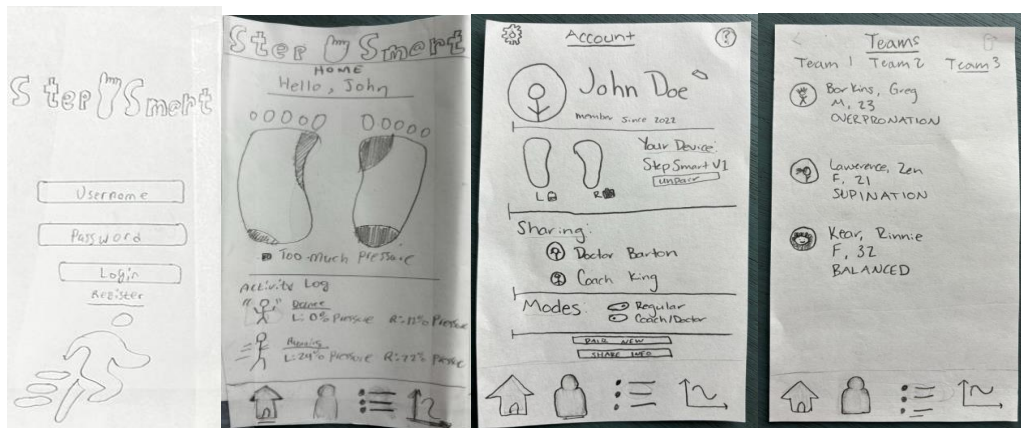
Insole Insert



Figure 7. Final Insert Paper Prototype

Similar to the initial prototype, there are different sizes for different types of shoes as well as sensors on the main part of the foot for ideal pressure tracking. They are still made with thin and flexible material to be comfortable and seamless in the shoe; however, there is now grip added to the underside of the insert to prevent movement in the shoe.

App



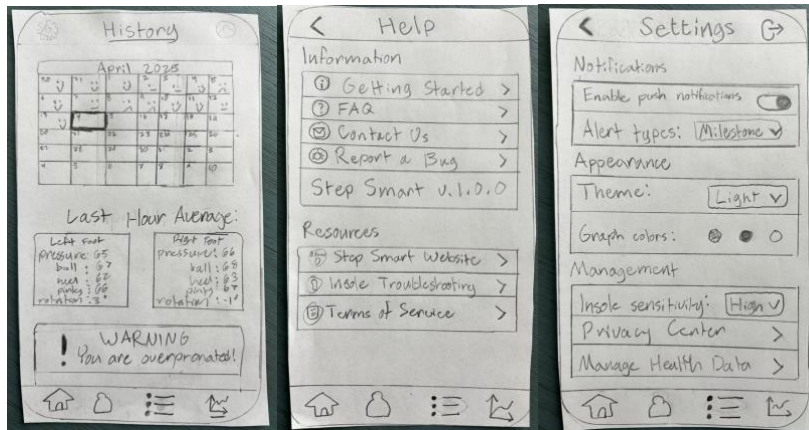


Figure 8. Final App Paper Prototype

Similar to the initial paper prototypes, the app includes a login screen as well as a home page. The home page shows real time positioning and the activity log. The account screen still shows the device list and sharing list, but now it offers a coach mode toggle, this will determine if you need a teams page. The teams page, which used to be the view page, allows the coach or instructor to view team members wearing the device as well as now switch between different teams. The history page still shows the users data over time, and highlights if the foot position was good or bad on specific days with smiley/sad faces. There is now a help screen to view important information and a settings page to change the look of the app and provide push notifications for real time data notifications.

Tasks

Know How One's Feet Are Positioned During Activities

- Similar to before, once the user is wearing the insole during their activity, they or their coach can open up the app to see where their pressure resides in their foot in real time. This allows them to correct their feet accordingly to prevent injuries. With the added grip, the tracking of pressure is more accurate as the insole stays in the correct place. The app's added intuitiveness allows for an easy view of the data as well.

Use an App to See Health and Fitness Data Visuals

- With improved data presentation from this app, users can see their data even better now. The history page now shows good/bad days in a user friendly visual way on the calendar. The live notifications allow for the user to get real time notifications if their foot is dangerously positioned. Also, The coach mode toggle allows for different users to have their correct form of experience. Overall the user experience has significantly

improved, which allows them even better access to tracking their data with all the original elements from before.

DIGITAL MOCKUP

Insole Insert

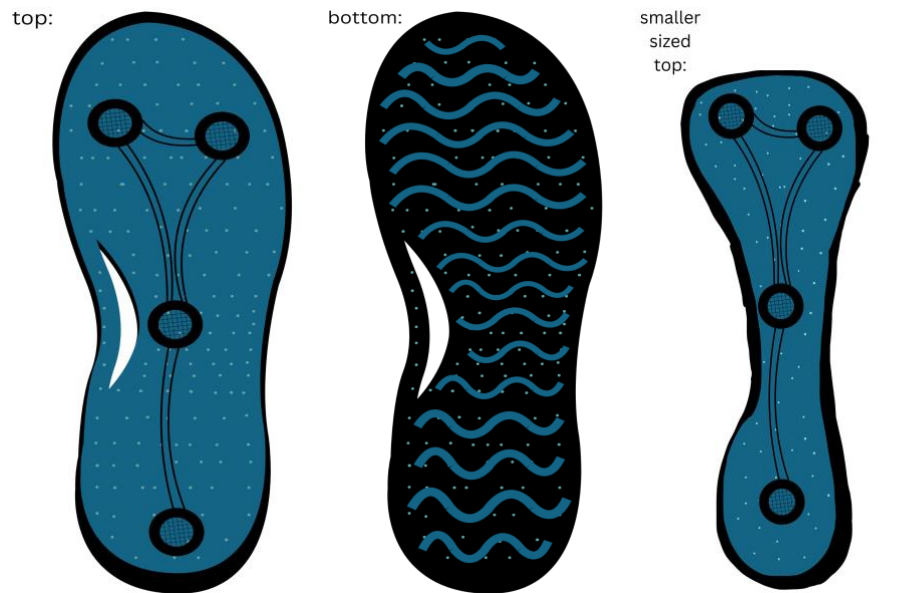


Figure 9. Insert Digital Mockup

Not a lot changed going from the final paper prototype to this digital mock up, but a new sensor on the arch was added. We found that different sizes and grip were well appreciated by the users, but in order to maximize pressure assessment to ensure proper calculation for overpronation and supination, we wanted to include a sensor on the arch of the foot.

App



Figure 10. App Digital Mockup

With the same screens as the final paper prototype, we improved the overall visual interest and flow of our design. We improved the bottom nav bar by using more intuitive icons and adding an underline to the icon corresponding to the specific page you are on. The top app bar was improved by adding a notifications button to all the screens for quick access to live notifications as well as moving the settings and help buttons to the account menu as that is the more instinctive place for them. The teams menu also has the option to add new members and teams and the login page has an option to “sign up” and “forgot password”.

When it came to switching from paper to digital, we found that the most significant revisions we made based on feedback from heuristic evaluations and usability testing were the updates to the top app bar and the view page. In our original prototype, the top navigation bar only displayed the title of the app, which left users confused about their current location within the app. It lacked essential navigation elements such as sign-in/sign-out options, back buttons, settings, or help, making the experience disorienting and inefficient. By redesigning this component to include key navigational buttons and clearer page indicators, users can now

more easily understand where they are and how to move through the app, significantly improving usability. Another critical revision was the redesign of the view page. Initially, this screen simply listed people and their data without clearly communicating its purpose or functionality. We also did not find good icons to use at first. Using first a magnifying glass, then an eye, we failed to convey that this page was intended for coaches or instructors to view team member information. We updated both the icon and the page title to better reflect its function. The icon is now a list, and the page is labeled “Team.” It now allows users to switch between different teams and view student data more intuitively. This change enhances clarity, aligns with user expectations, and supports the app’s core purpose more effectively.

Tasks

Know How One’s Feet Are Positioned During Activities

- As before, the insert is connected to the app to allow for real time data synchronization so as the user is wearing the insert, they can view the app to see their foot position. With the new added sensor, tracking data is even more precise as overpronation/supination is more accurately calculated. The notifications button that was added to all the pages app bar allows for quicker access to receive the important notifications about one’s foot. Again, the overall improvement of the app’s interface makes it much easier and user friendly for people trying to view how their feet are positioned during sports.

Use an App to See Health and Fitness Data Visuals

- The cleaner interface with consistent styling and color scheme make it much more pleasing to the eye to view one’s data. The user experience has been improved significantly by making the app more streamlined while still keeping all the necessary functionalities and graphs to clearly show foot posture.

DISCUSSION

Our main takeaway from this design journey is that creating something from scratch is a long and complex process with a mountain of ideas, revisions, and feedback. The first design is never going to be the final product, and by the end it will have been transformed through feedback and revisions into a new form entirely. Multiple series of interviews and testing need to be done to get a well rounded opinion of product usability. Multiple people with different backgrounds need to be assessed to see the picture from all sides. You need to be ready to listen to all critiques and advice with open ears and try to walk in your user's shoes to find what works best for your product.

The process this semester didn't result in a fully reinvented design, rather it resulted in a fully realized design. The different forms of testing, revisions, and analysis helped us polish our concept from a good idea to a full-fledged product with solid and sound design considerations. Initially, we were unsure of what our product actually was. Everyone had a different version in their mind of what it would look like and do, and through collaboration we eventually settled on a design for the insole and app we were all satisfied with. The group feedback sessions forced us to truly form our design into something that wasn't just possible, but well thought out and useful.

Our tasks stayed the same throughout the usability tests. Since the beginning our main tasks were to allow users to know how their feet are positioned during their activities (which was informed by our initial idea) and to use an app to see health and fitness data visuals, which came naturally when we needed a way to see the data our insole was recording.

If we intended for this product to go to market, we would have had many more iterations of our design, but as this semester was just conceptualization, we believe that we're at the limit of improvement for our prototypes. Since the furthest we've taken our design is a non-interactive digital mockup, the feedback we can receive is limited to what our test participants can imagine using the product is like. Just a piece of hardware such as our insole alone would take dozens of iterations to get right, and that's not even considering the software integration.

Overall, we are very pleased with our design and we greatly value what we've learned in the process of creating and refining it. It's been a long but fun process that has required us to employ all aspects of our creativity.

APPENDIX

Design Critique Results

Group 5: Adrien, Daniel, Kehajana, Jonathan

- Would we be interviewing people with joint injuries specifically?
- How would tracking data be used?
- Different kinds of shoes so we want to make sure it works for as many people as possible – good question to ask what type of shoe they use
- Interviewing college student (18-22 year old), dance instructor (25-30 year old), boss that runs (50s) – good range of people
- How would an app for coaches and instructors work for all students to be tracked?

Group 2: Jacob, Chen, Jo

- Agreed with experience discomfort
- And how wearing it would feel since its a foreign object – pebble or paper
- Price point – subscription?
- Waterproof
- Pressure sensor thinness
- Pressure sensor key points – ball, heel, arch, other side of ball
 - Ballet shoes are just made out of canvas with split soal usually (and jazz shoes)
 - Different types of models for different activities
- Would we rather have a flexible sensor model that can fit different types of activities or specified models for individual activities and need multiple?

Design Research Questions

1. What is your history of ankle pain/injury?
2. What type of shoe/product have you used for your activity?
3. What type and how often do you participate in your activity?
4. Would you prefer insoles or clip-on sensors?
5. Do you stretch/prep before the activity to prevent injury?
6. What they would want out of a data tracking app?
7. What type of price point would you want for a product like this?
8. Would you like to be able to share your data (doctor/coach)?

Heuristic Testing Results

Groups that evaluated us and the ratings they gave us:

- Group #5 – Adrien, Daniel, Kehajana, Jonathan
 - [H3 User control and freedom] [Severity 3]

- [H10 Help and documentation] [Severity 4]
- All other heuristics were 0
- Group #4 - Beau, Alexis, Dalton
 - [H4 Consistency and standards] [Severity 1]
 - [H10 Help and documentation] [Severity 1]
 - All other heuristics were 0

Groups that we evaluated:

- Group #5 – Adrien, Daniel, Kehajana, Jonathan
- Group #4 - Beau, Alexis, Dalton

Notes:

- From group 5:
 - Add buttons on app bar like settings, login/logout, back buttons
 - Add page to show how to link to the doctors for sharing
 - Add page to show intermediate connecting screens/modals
 - Add help page
 - make app bar show what page it is currently on
 - “add user” button in view page
- From group 4:
 - Show a form of feedback for a sign of approval on the calendar for days that have good positioning like a checkmark and on really bad days show a red X
 - Add notifications for when position is bad
 - Add arch sensor or more sensors overall
 - Make this just go underneath the existing insole so its protected
 - Add scroll bar for app in view page
 - Possibly add reward system like the apple fitness rings, more clear icons

Usability Testing Results

Critical Incidents:

- Insert was “slippery” (obviously made of paper but not clearly shown)
- App bar displaying app name rather than page name was confusing
- “View” page is only necessary for coaches/instructors and would not really want it showing up on her app as an unnecessary button.
- View page was unclear as to what it’s used for
- Hard to find settings and help page

Contributions:

Jayme Greer: 70% – helped write this document, contributed to prototyping and testing

Evan Bohler: 20% – helped write this document, contributed to prototyping and testing

Sophia Miller: 5% – contributed to prototyping and testing

Charlie Whittleman: 5% - contributed to prototyping and testing